Benchmarking and improving imputation approaches for recurrent inversions in the human genome

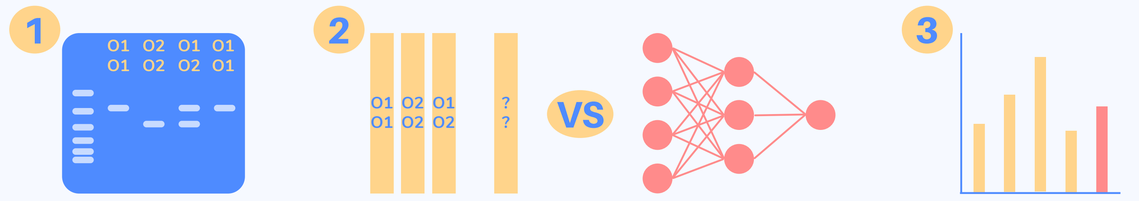
Illya Yakymenko[1], Jon Lerga-Jaso[1], Mario Cáceres[1,2]

[1]Institut de Biotecnologia i de Biomedicina, Universitat Autònoma de Barcelona, Bellaterra (Barcelona), Spain; [2]ICREA, Barcelona, Spain

Motivation

- Inversions flanked by inverted repeats (IRs) tend to be recurrent.
- Due to recurrence, inversions usually show low LD with nearby variants ($r^2 \ll 1$).
- The effect of recurrent inversions is probably missed in association studies.

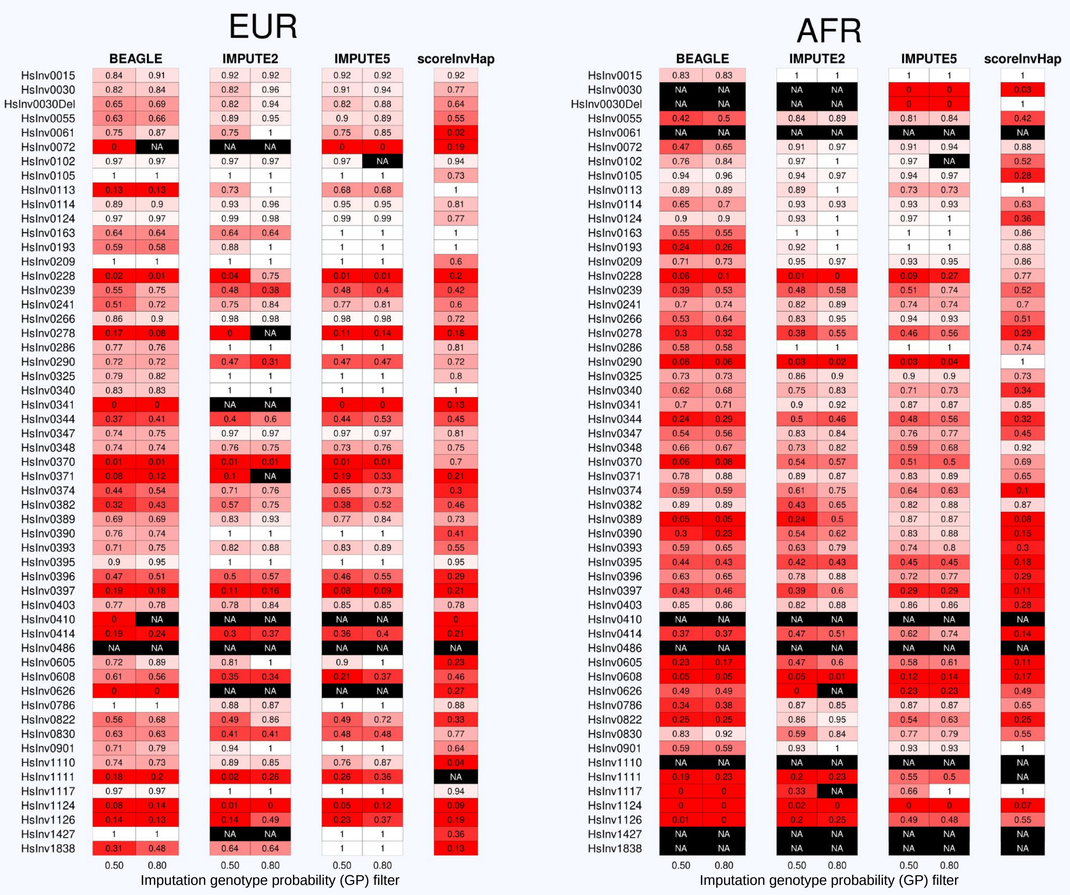
Procedure



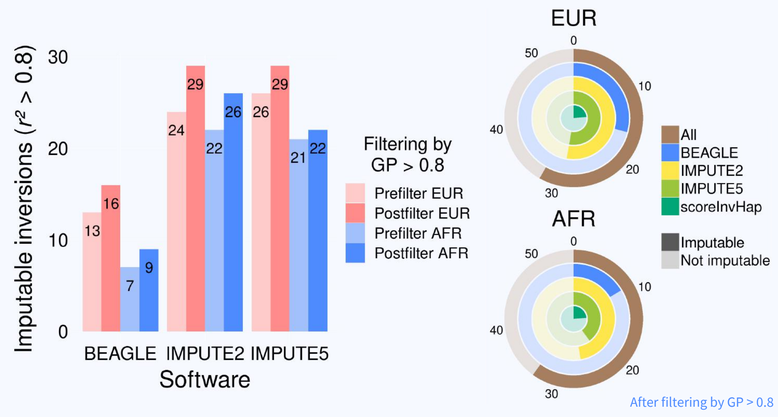
1. Experimental genotyping (95-550 1KGP samples)
2. State-of-art imputation
3. Deep Learning imputation
3. Benchmarking

Comparison between imputation algorithms

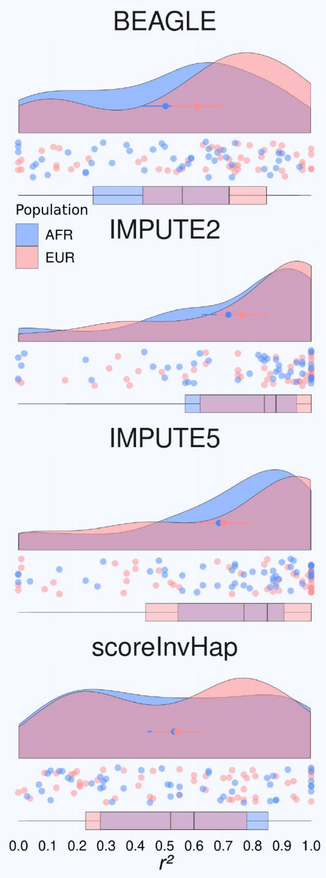
Inversion imputation accuracy (r^2) of different algorithms



Inversion imputability (r^2> 0.8)

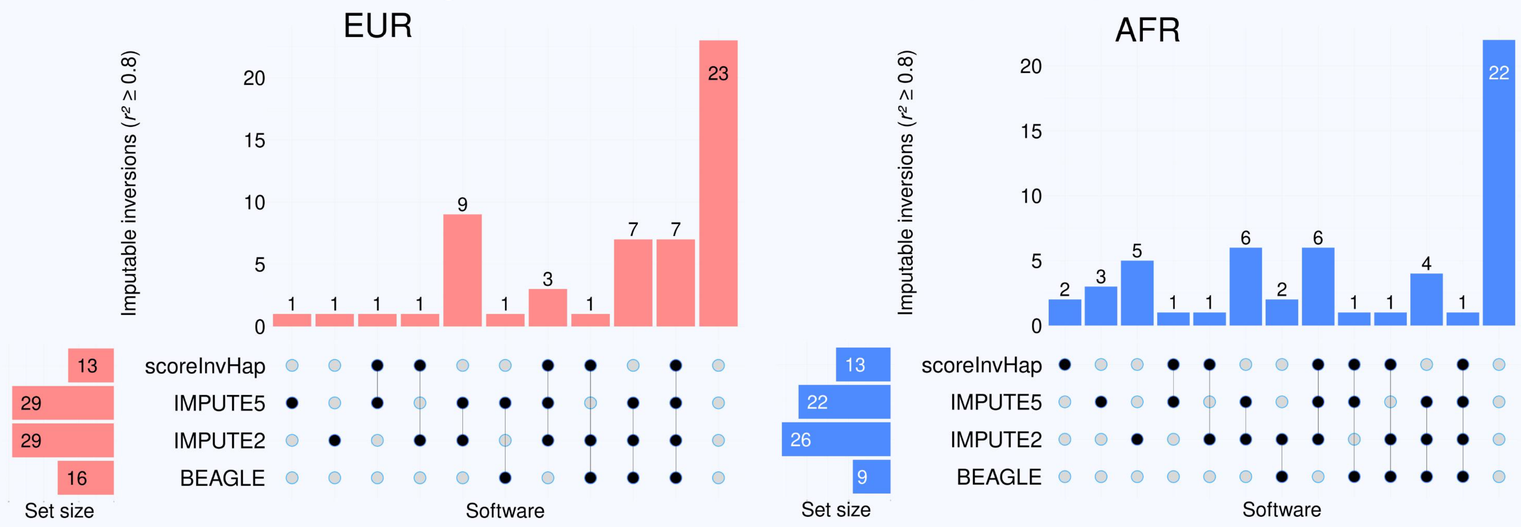


Distribution of imputation accuracy



- Filtering by genotype probability (GP) improves imputation r^2 value at the expense of filtering out samples with low certainty.
- European samples are more imputable than African samples.
- IMPUTE2 and IMPUTE5 show the best imputation performance.
- Around half of recurrent inversions (32/55 for EUR, 33/55 for AFR; both GP > 0.8) are imputable (r^2> 0.8).

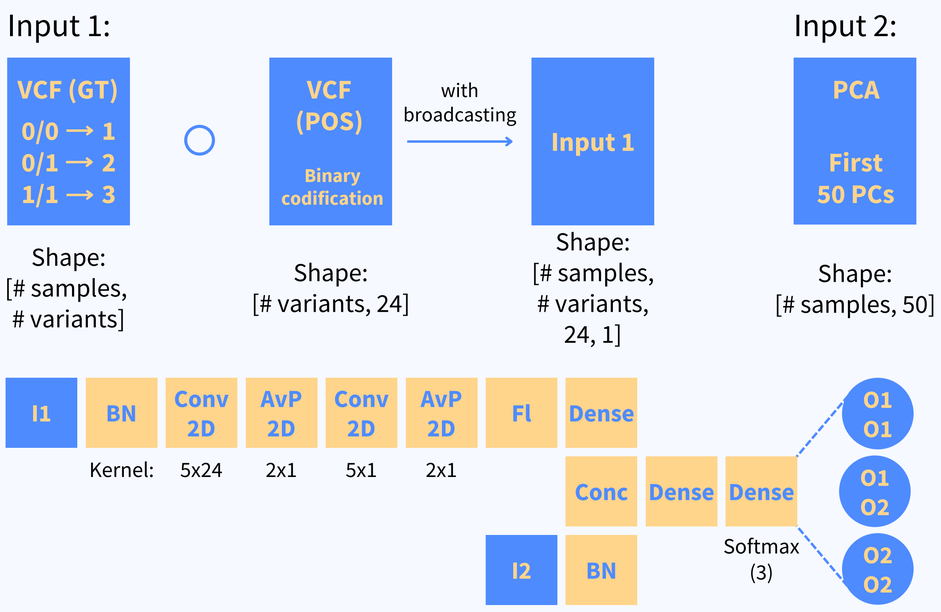
Algorithm contribution to inversion imputability



- EUR and AFR populations have different imputability profiles:
  - European samples are more imputable by multiple algorithms.
  - Several inversions are only imputable by one of the programs in African population.

Deep Learning implementation and performance

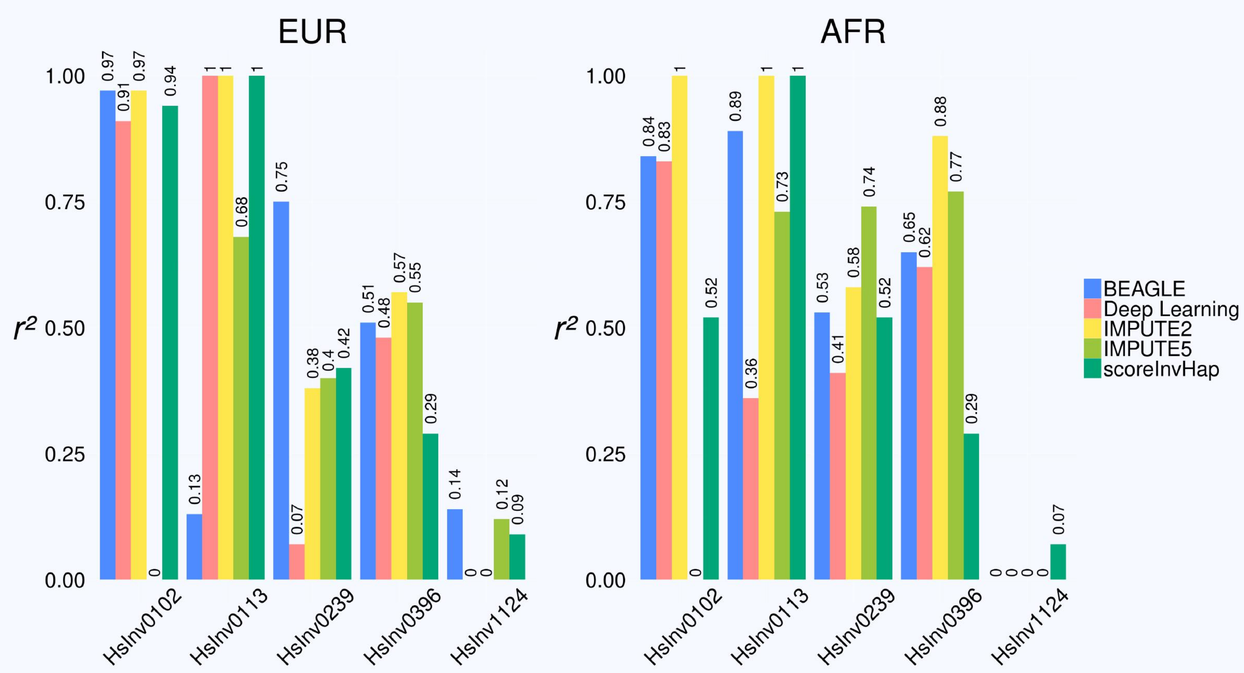
Model



- During training the learning rate is reevaluated every 20 epochs.
  - If training loss fluctuation is high, the learning rate is diminished.
  - If training is slow and without fluctuation, the learning rate is augmented.
- Despite the raise in validation loss during training, validation accuracy remains stable.
- Training loss minimum value depend on inversion characteristics. Less imputable inversions tend to have greater minima.

Preliminary results

Comparison between state-of-art and Deep Learning algorithms



- The Deep Learning approach shows similar performance in a small subset of inversions with different degrees of imputability, but it still needs improvements.
- The model does not distinguish well the populations from the Principal Component input. The presence of different populations in the reference might be hampering the imputation due to possible overfitting to haplotypes of a specific population. This may explain the low performance for HsInv0113 in AFR population.

Contact: illya.yakymenko@uab.cat | mcaceres@icrea.cat — https://grupsderecerca.uab.cat/cacereslab
Funding: Research grant no. PID2019-107836RB-I00 and Predoctoral Contract PRE-2020-092440.